

Curriculum Vitae

Mahyar Karimi

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Contact Details

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Education

2023 – **M.Sc. in Computer Science**,
2026 Institute of Science and Technology, Klosterneuburg, Austria.
 Advisor: Prof. Thomas Henzinger
 Expected date of graduation: March 31, 2026

2018 – **B.Sc. in Computer Engineering**,
2023 University of Tehran, Tehran, Iran.
 Graduated as one of the top 10 students in my class.
 Thesis advisor: [Prof. Hossein Hojjat](#)

Notable Coursework (full mark is 20/20):

Software testing:	19.4,	Database design:	19.1,
Distributed systems:	17.8,	Cyber-physical systems:	16.3,
Software engineering:	17.8,	Internet engineering:	18.4,
Computer architecture:	20.0,	Operating systems:	19.5.

Research Experience

06.2023 – **Research intern**, Institute of Science and Technology Austria (ISTA)
09.2023 Worked under the supervision of Prof. Thomas Henzinger.

We studied how neural networks can act as proofs. For this project, I wrote automations for network training and assembling (networks of networks) using PyTorch, translating networks to SymPy trees and SMT-LIB queries, and passing these queries to SMT solvers (Z3 as a generic and Marabou as a specialised solver). I also modified Marabou to fine-tune its abstract interpreter.

07.2022 – **Research intern**, Institute of Science and Technology Austria (ISTA)
09.2022 Worked under the supervision of Prof. Thomas Henzinger.

We designed monitors for quantitative fairness of online decision-making agents. We modelled agent-environment interaction as a Markov chain. I wrote a Rust prototype to generate monitors and evaluate their performance. To improve space usage, I also wrote a parametrised Markov chain that generates states on-the-fly.

2021 – **Research assistant**, University of Tehran
2023 Collaborated with [Prof. Hossein Hojjat](#) and [Amir Hossein Seyhani](#).
We developed a model for causal reasoning over concurrent systems, with Winskel's *event structures* as our concurrency model, and the Halpern-Pearl definition of root cause. We have a Python tool for causality checking, with significant optimisations for verifying a candidate root cause.

Industry Experience

2022–3 **Software engineer**, [Divar](#).
I worked in the *customer trust* team, where we found ways to find and stop fraudulent users by tracing their behaviour across the platform. Before that, I worked in the *jobs* team, a new direction of the company for making a job market functionality.
Related skills: Microservices architecture, Django, CI/CD, Scrum, Kanban

2021 **Infrastructure intern**, [Tapsell](#).
Implemented a simple query caching mechanism, with Redis as the cache.
Related skills: Kubernetes, Redis, Nginx, Docker

Teaching Experience

2020 – University of Tehran
2022 Teaching assistant in the following courses:

- *Advanced Programming* (Lecturer: [Prof. Ramtin Khosravi](#))
For this course, the TAs would work in teams for designing assignments. Students would receive a review of their code and oral feedback from the TA during grading.
- *Data Structures* (Lecturer: [Prof. Fathiyeh Faghieh](#))

Extracurricular Coursework

05.2023 **Functional Program Design in Scala**
Lecturer: Prof. Martin Odersky (EPFL).

03.2023 **Functional Programming Principles in Scala**
Lecturer: Prof. Martin Odersky (EPFL).

11.2023 **Supervised Machine Learning: Regression and Classification**
Lecturer: Prof. Andrew Ng (course offered by DeepLearning.AI).

Awards & Honors

2022 Research scholarship, granted by the Austrian Agency for Education and Internationalisation ([OeAD-GmbH](#)).

Presentations

- 09.2023 [Monitoring Markov Chains](#)
Presented at the 15th Alpine Verification Meeting, Prague, Czechia.
- 09.2024 [Monitors, Systems, and Privacy](#)
Presented at the 16th Alpine Verification Meeting, Freiburg, Germany.

Academic Service

- 2025 **Artifact evaluation committee member**, CAV 2025 (International Conference on Computer Aided Verification).
- 2025 **Subreviewer**, ECAI 2025 (European Conference on Artificial Intelligence) and FMCAD 2025 (Formal Methods in Computer-Aided Design).

Publications

Refereed Conference Papers

- [1] Thomas A. Henzinger, Mahyar Karimi, and K. S. Thejaswini. “Privacy-Preserving Runtime Verification”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. CCS ’25. Taipei, Taiwan: Association for Computing Machinery, 2025, pp. 2774–2787. ISBN: 9798400715259. DOI: [10.1145/3719027.3765137](#).
- [2] Thomas Henzinger et al. “Runtime Monitoring of Dynamic Fairness Properties”. In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. FAccT ’23. Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 604–614. ISBN: 9798400701924. DOI: [10.1145/3593013.3594028](#).
- [3] Thomas A. Henzinger et al. “Monitoring Algorithmic Fairness”. In: *Computer Aided Verification*. Ed. by Constantin Enea and Akash Lal. Cham: Springer Nature Switzerland, 2023, pp. 358–382. ISBN: 978-3-031-37703-7.